Adjustable mandrel can be extended to exactly fit work in hand

spindle with almost no vibration and run-out, measured with a dial gauge, of less than 0.05mm. As a result, the lathe runs smoothly and quietly and sits solidly on its rubber mounting feet without the need for bolting down, having no tendency to creep across the bench when running.

The 375W DC brush motor is rated for continuous running and controlled by electronics that allow speed to be varied from 400–3600 rpm. The power produced is all that would be expected from a ½ HP motor, which is more than adequate for the work that will be required of it.

A unique characteristic of the Precision Pro is the provision of an ER20 collet chuck built into the spindle in place of the more normal Morse taper, the tailstock being 1MT. This is an excellent choice not only for mounting pen mandrels but also for holding small components for precision work such as thread cutting operations when making bespoke or kitless pens. A collet chuck guarantees concentricity and as a result, the run out at the end of a fully extended mandrel remains less than 0.05mm even without tailstock support. Tailstock support is of course provided and takes the form of a hollow tailstock drive centre through which the mandrel rod fits with no play, thus virtually eliminating 'out of round' issues caused by non-concentric running.

The mandrel and hollow centre are supplied with the lathe along with basic 7mm 'slimline' type bushes. Also supplied are a collet spanner, 80 and 150mm toolrests and a centre knock-out bar. The toolrest sizes are ideal for single barrelled or two-part pens.

Available as optional extras are a light pull drive, polishing mop arbor, 10mm and 16mm four prong drive centres, 38mm woodscrew chuck and 50mm faceplate. This extends the versatility of the lathe beyond the pen maker to those who need to undertake small spindle or faceplate work to a high standard but do not have the space or the need for a full size lathe. Mounting the drive centres etc. in the collet chuck drive will require the purchase of an additional 12mm collet.

The one option that is not currently available, but which I believe would extend the versatility of this machine even further is that of a small scroll chuck to fit the non-standard 24 x 1mm spindle threads. Perhaps an additional thread option could be considered for the Clubman SK80 chuck. This would not only increase the options for small bowls and forms but also enable drilling of pen blanks larger than 13mm diameter to be carried out on the lathe with the provision of suitable 1MT tailstock chuck.

Conclusion

The Axminster Trade Series Precision Pro Lathe fulfils or exceeds all of the requirements that I set out at the beginning of this review and will meet the needs of many turners, both serious amateur and professional, whether specialising in pen making or small spindle work. The optional extras extend its versatility to small faceplate work. If the option of a small scroll chuck were to be made available, its utility would be extended even further, for example to the small cabinetmaking business which requires a small lathe for precision turning of handles and small spindles, etc. but cannot justify the expense or space requirements of a full size lathe. For those requiring a professional lathe that is small and portable yet robust and accurate I thoroughly recommend it.

PROS:
- Professional quality in a small lathe
- Electronic variable speed
- Portability
- Accuracy

CONS:
- Expensive
- Lack of a scroll chuck option

I PARTICULARLY LIKED:
- The collet chuck and hollow tailstock mandrel support
- Quiet running and lack of vibration

Optional drives with supplied tailstock centre and toolrests

Contact: Axminster Tools & Machinery **Tel:** 0800 371 822 **Web:** www.axminster.co.uk

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